A COMIC ABOUT ARTIFICIAL INTELLIGENCE

QAI-QALA

Qadam and the Ten Days Before the Blizzard

QAZAQ · AI · QALA — “qala” is Kazakh for **city**

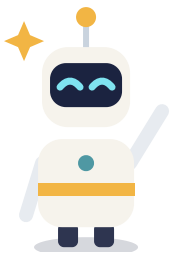


Play the game
scan to open QAI-QALA

AIF1303 AI Fundamentals · Qairu University
Team 11 · Yuldashov Bilal · Tashmetov Shakhrukh
10 lectures · 10 days · 10 districts

MEET THE CAST

Ten residents, one robot, ten days



THE HERO

Qadam

A small city robot. “Qadam” is Kazakh for **step** — he learns one AI idea per day, step by step, and every idea powers a new system of the city.

AND YOU — THE CITY ENGINEER

A huge blizzard (**боран**, *boran*) reaches Astana in ten days. Each chapter is one day of the countdown and one week of the course. Read the story, study “Qadam explains”, then scan the QR code and solve the same task in the game — that is how district after district lights up.



Assel
meteorologist
Day 10 · Ch. 1



Aigerim
hospital doctor
Day 9 · Ch. 2



Nurlan
site foreman
Day 8 · Ch. 3



Serik ata
toguz kumalak master
Day 7 · Ch. 4



Dana
traffic inspector
Day 6 · Ch. 5



Yerlan
school principal
Day 5 · Ch. 6



Marat
bus driver
Day 4 · Ch. 7



Zhanna
akimat secretary
Day 3 · Ch. 8



Timur
power-grid engineer
Day 2 · Ch. 9



Aliya
schoolgirl
Day 1 · Ch. 10

HOW TO READ THIS COMIC

- **Left page** — the story of the day. The big number is the countdown to the blizzard.
- **Right page** — Qadam explains: key ideas, a diagram, a worked example, a myth, a Kazakh word, a quiz.
- **QR codes** open the same chapter inside the QAI-QALA game, so the comic and the game are one book.
- Every number in the comic is the same as in the game, and every result was recomputed by our code.

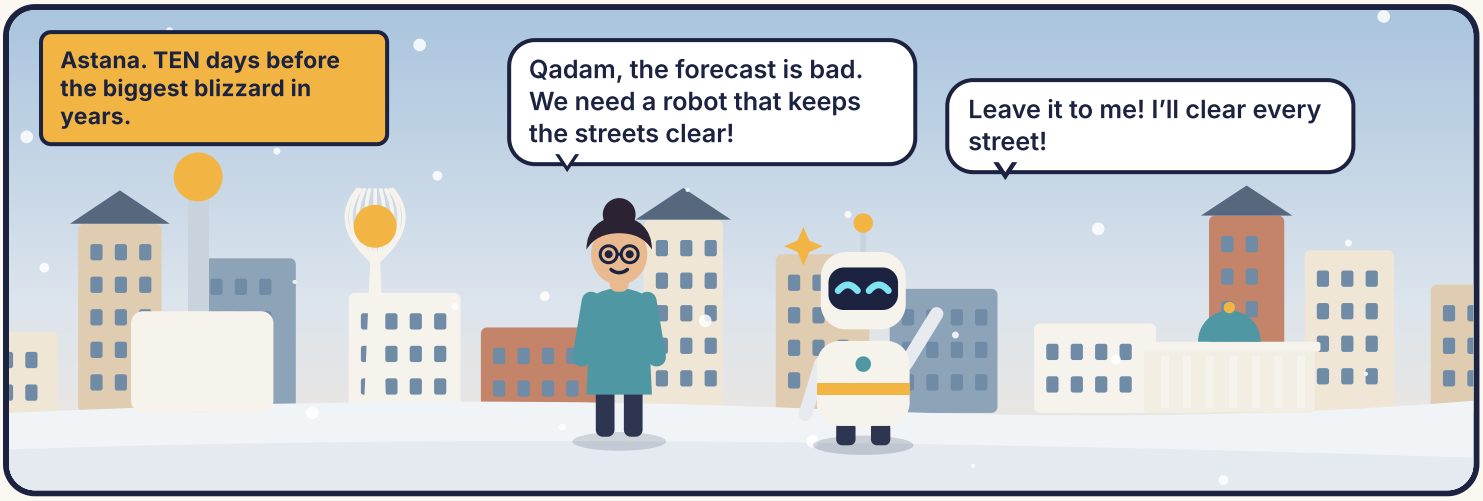
10
DAYS LEFT

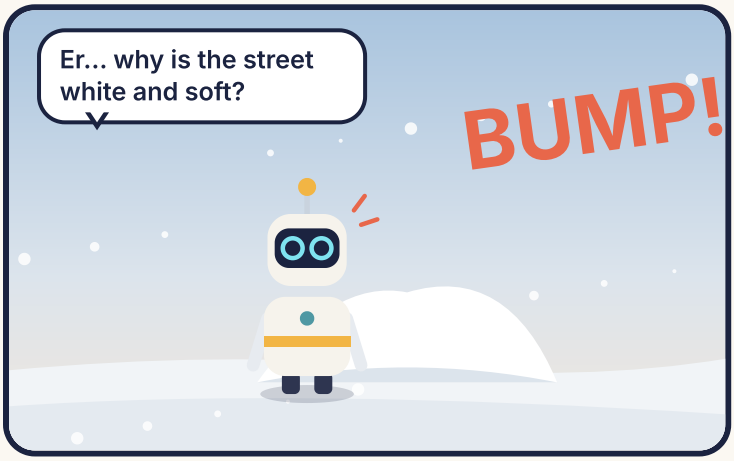
CHAPTER 1 · ASSEL, METEOROLOGIST

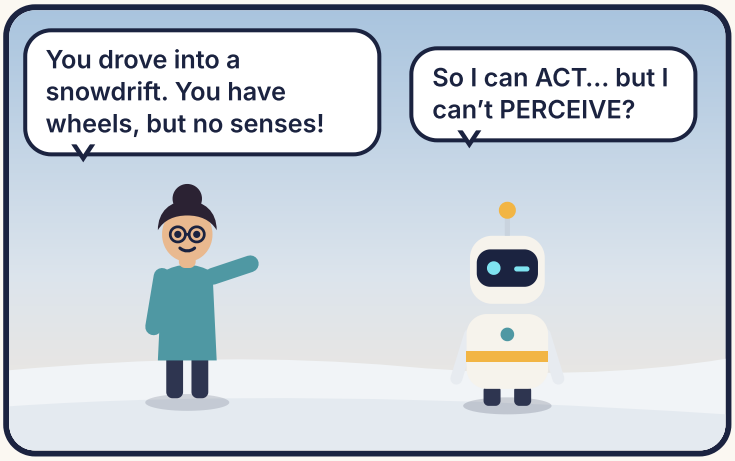
The Robot Who Couldn't See

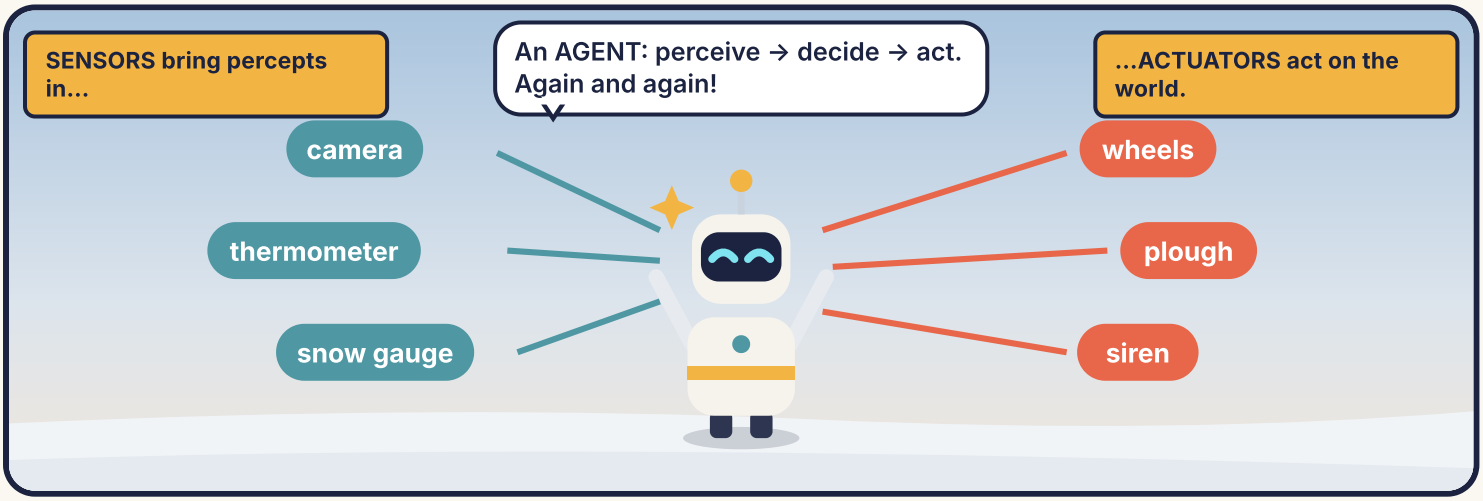
Week 1 · Intelligent agents & environments

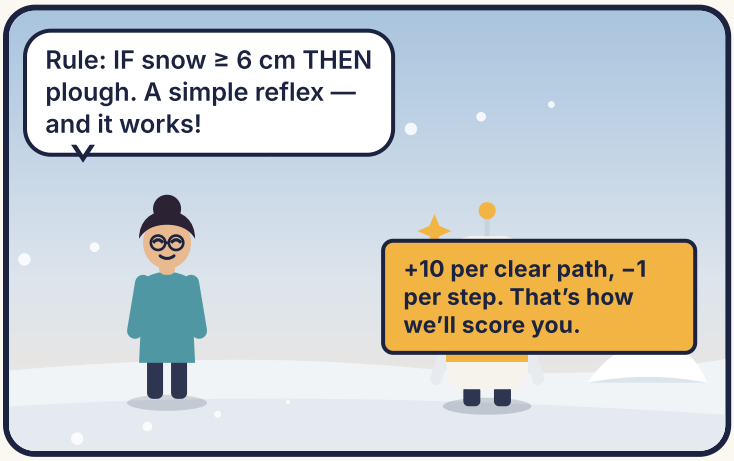
Play it
Open this chapter in QAI-QALA











WHAT QADAM LEARNED TODAY

An agent perceives with sensors and acts with actuators. A rational agent picks the action that is expected to maximise its performance measure.



"Now Qadam notices every drift and warns people himself."

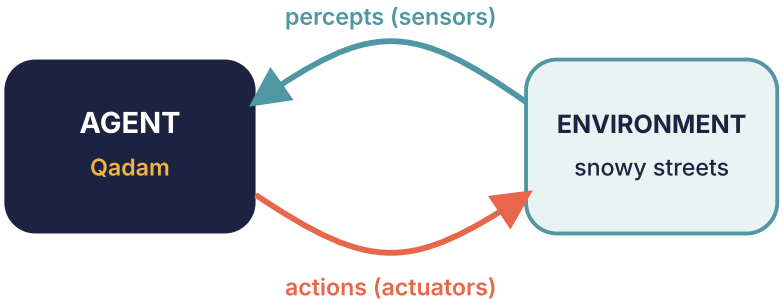
QADAM EXPLAINS · CHAPTER 1

Intelligent agents & environments

AI in this course = building RATIONAL AGENTS (Russell & Norvig): systems that act to achieve the best expected outcome.

KEY IDEAS

Agent	Anything that perceives its environment through sensors and acts on it through actuators.
Agent function	Maps the percept sequence (everything seen so far) to an action.
Rational	Chooses the action expected to maximise the performance measure, given its percepts and knowledge. Rational ≠ all-knowing.



The agent loop: percepts in, actions out.

P · performance	E · environment	A · actuators	S · sensors
clear paths, safety, few steps	snowy streets, people, cars	wheels, plough, siren, display	camera, thermometer, snow gauge

PEAS of Qadam the snow robot

FIVE KINDS OF AGENT

- **Simple reflex**: IF snow ≥ 6 cm THEN plough
- **Model-based**: remembers streets it can't see right now
- **Goal-based**: plans a route to the hospital
- **Utility-based**: trades speed against safety
- **Learning**: gets better from experience

WORKED EXAMPLE

Qadam cleared **4** paths in **15** steps. Score = 10 × 4 − 15 = **25**.

PROPERTIES OF THE ENVIRONMENT

Qadam's world is **partially observable** (snow hides kerbs), **stochastic**, **sequential**, **dynamic**, **continuous** and **multi-agent** — the hard side of every property.

QADAM'S GLITCH · MYTH VS FACT

"A rational agent never makes mistakes."

Myth. It picks the best action *expected* from what it knows. With missing information it can still be unlucky.

KAZAKH WORD OF THE DAY

Ауа райы

aua raiy — weather

DID YOU KNOW?

The term "artificial intelligence" comes from the 1955 proposal for the Dartmouth summer workshop of 1956 (McCarthy, Minsky, Rochester, Shannon).

CHECK YOURSELF

1. Which PEAS letter is the snow gauge?
2. Is "IF snow ≥ 6 cm THEN plough" a reflex or a goal-based agent?
3. Qadam cleared 6 paths in 25 steps. What is his score?

Answers on page 24.

Try it in QAI-QALA. District 1: solve its task with fresh numbers every time — guessing won't work.

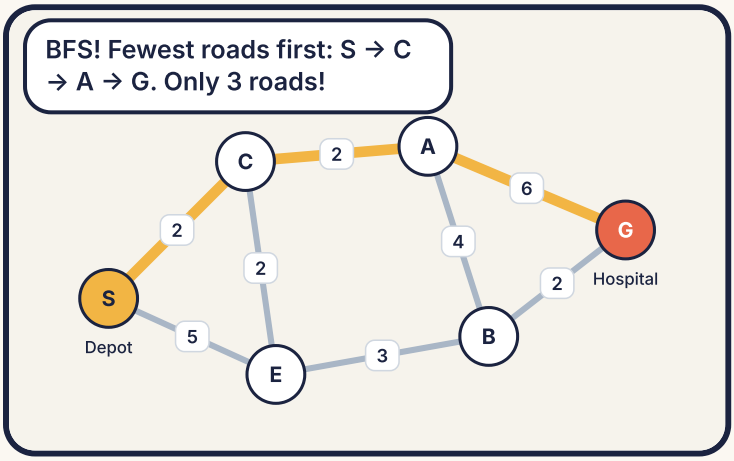
CHAPTER 2 · AIGERIM, HOSPITAL DOCTOR

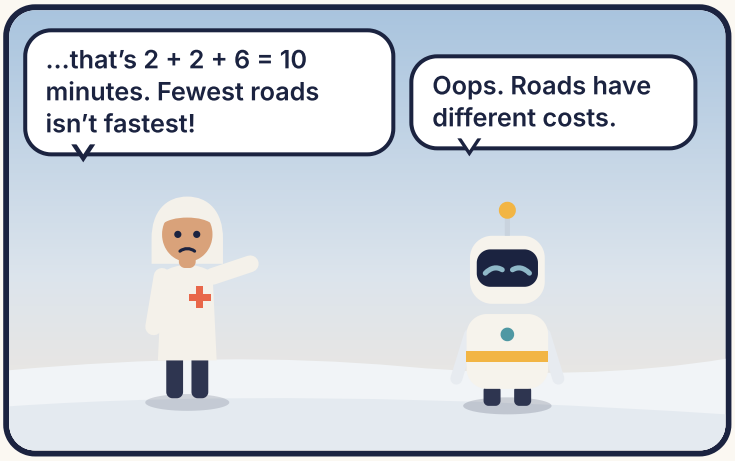
Medicine Before the Storm

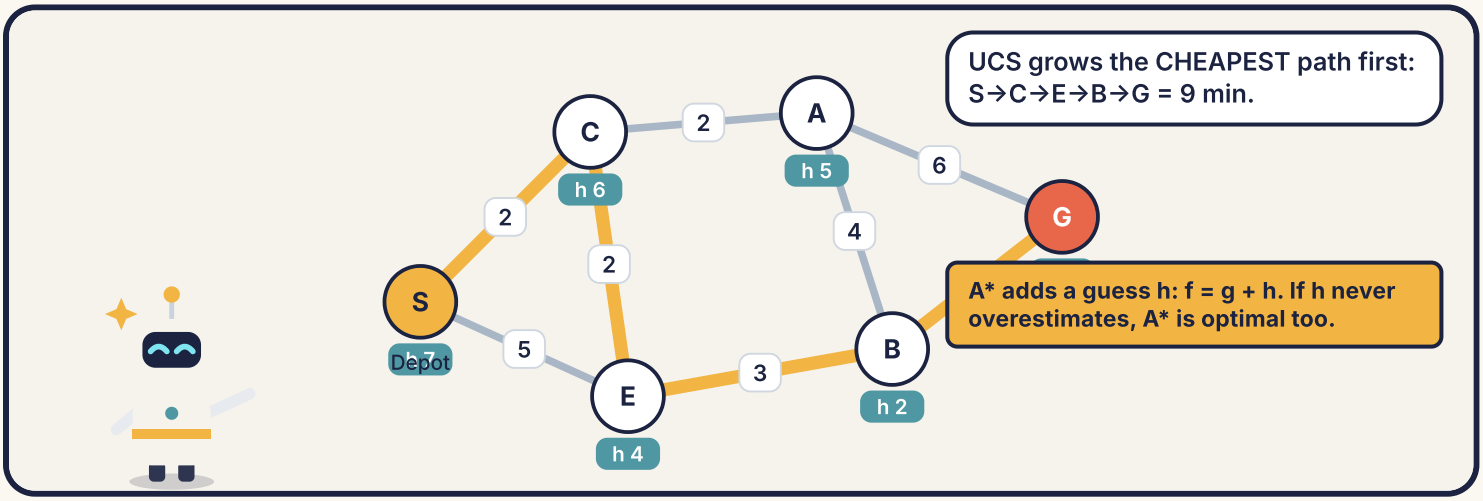
Week 2 · Uninformed search, A* & heuristics

Play it
Open this chapter in QAI-QALA











WHAT QADAM LEARNED TODAY

Search algorithms differ only in WHICH path they extend next. UCS and A* (with an admissible heuristic) find the cheapest route.



"The medicine arrived in 9 minutes. You may have saved someone tonight."

QADAM EXPLAINS · CHAPTER 2

Uninformed search, A* & heuristics

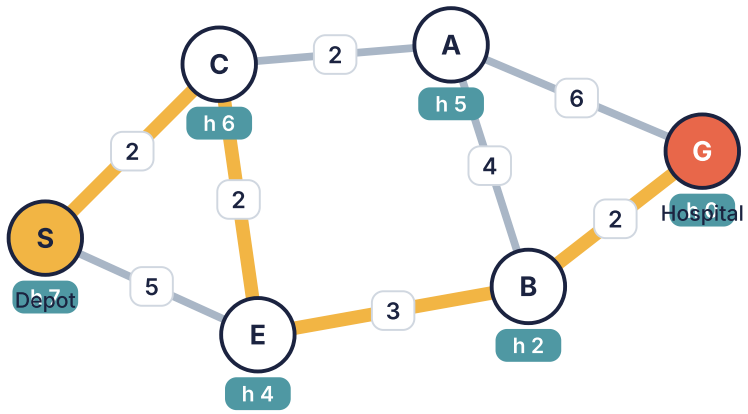
A search problem: states, actions, a transition model, a goal test and a path cost. Algorithms grow a tree of paths from the start.

KEY IDEAS

Frontier — The paths waiting to be extended. BFS: FIFO queue. DFS: LIFO stack. UCS & A*: priority queue.

Heuristic h(n) — A guess of the cost still left. Admissible = never overestimates.

A* — Extends the path with the smallest f = g + h. Optimal when h is admissible.



Our map: minutes on roads, teal = h(n). Gold = cheapest route.

Algorithm	Extends first	Route on our map	Cost
BFS	fewest roads	S→C→A→G	10
DFS	deepest path	S→C→A→G	10
Greedy	smallest h	S→E→B→G	10
UCS	smallest g	S→C→E→B→G	9
A*	smallest g + h	S→C→E→B→G	9

Computed on the same map as the game

WORKED EXAMPLE

A* from S: C has f = 2 + 6 = **8**, E has f = 5 + 4 = **9** → extend C first. Every h here is ≤ the true cost (e.g. h(E) = 4 ≤ 5), so h is admissible.

QADAM'S GLITCH · MYTH VS FACT

"A bigger heuristic is always better."

Myth. If h overestimates, A* can skip the best route. The best h is as big as possible *without* overestimating.

KAZAKH WORD OF THE DAY

Жол

jol — road

DID YOU KNOW?

A* was published in 1968 by Hart, Nilsson and Raphael while building Shakey, one of the first mobile robots.

CHECK YOURSELF

1. Which data structure holds the BFS frontier?
2. Which algorithm ignores road costs when it chooses?
3. True cost from E to the hospital is 5. Is h(E) = 4 admissible?

Answers on page 24.

Try it in QAI-QALA. District 2: solve its task with fresh numbers every time — guessing won't work.

8
DAYS LEFT

CHAPTER 3 · NURLAN, SITE FOREMAN

Homes and Bus Lines

Week 3 · Local search & constraint satisfaction

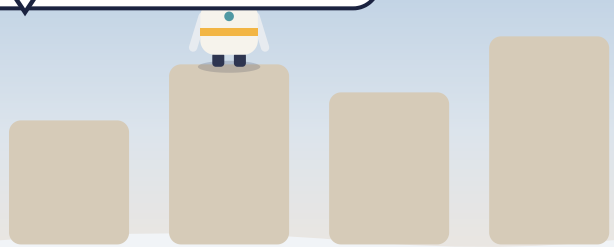
Play it
Open this chapter in QAI-QALA

Day 8. A building site.

Eight plots. Higher number = better plot. Find the best one for new families!



Always step uphill: 3 → 5... both neighbours are lower. Is 5 the top?

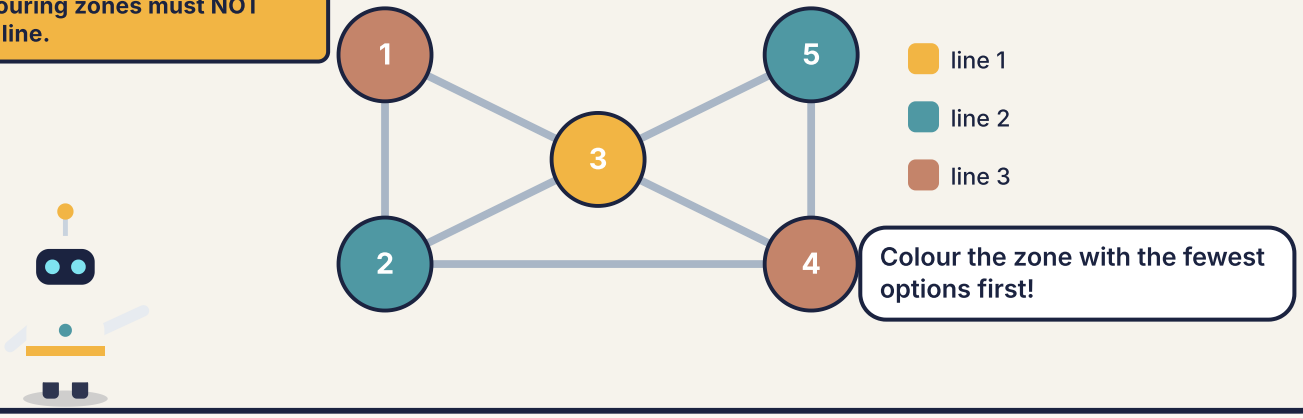


No! Plot 8 has 9. You're stuck on a LOCAL maximum.

Then sometimes I must step DOWN on purpose?



Next: 5 zones, 3 bus lines. Neighbouring zones must NOT share a line.



Colour the zone with the fewest options first!

Block built on the best plot. And no two buses fight for one road!



WHAT QADAM LEARNED TODAY

Local search keeps one state and improves it; it can get stuck on local maxima. A CSP finds values that satisfy every constraint.

"The block stands, and no two buses fight for one road."

QADAM EXPLAINS · CHAPTER 3

Local search & constraint satisfaction

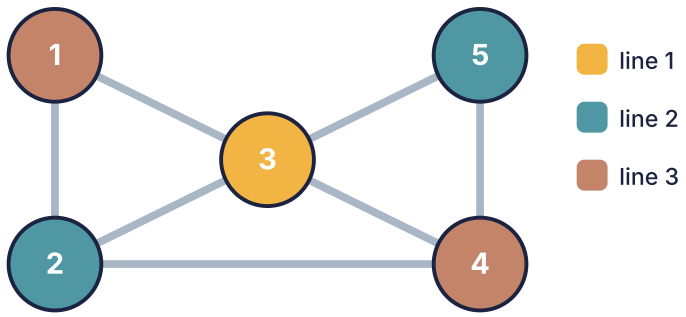
When only the final STATE matters (not the path to it), keep one state and keep improving it.

KEY IDEAS

Hill climbing	Move to the best neighbour. Stops at local maxima, plateaus and ridges.
Simulated annealing	Sometimes accept a worse move with probability e^(ΔE/T). High temperature T = brave, low T = careful.
CSP	Variables, domains, constraints. Backtracking + MRV (fewest values left) + forward checking or arc consistency.



Plots 3 5 4 6 8 7 5 9: from plot 1 hill climbing stops at 5.



One solution: zones 1 & 4 → line 3, 2 & 5 → line 2, 3 → line 1.

WORKED EXAMPLE

Annealing, a move worse by 2 (ΔE = −2): at T = 4 accept with e^(−0.5) ≈ **0.61**; at T = 1 only e^(−2) ≈ **0.14**.

QADAM'S GLITCH · MYTH VS FACT

"Hill climbing always finds the highest point."

Myth. It only sees neighbours. Random restarts or annealing help it escape local maxima.

KAZAKH WORD OF THE DAY

Үй

üi — house, home

DID YOU KNOW?

Sudoku is a CSP: 81 variables, domain {1...9}, 27 all-different constraints. Simulated annealing was introduced by Kirkpatrick, Gelatt and Vecchi in 1983.

CHECK YOURSELF

1. Values 2, 6, 4, 9. Hill climbing starts at 2. Where does it stop?
2. Name the three parts of a CSP.
3. What does the MRV heuristic choose first?

Answers on page 24.

Try it in QAI-QALA. District 3: solve its task with fresh numbers every time — guessing won't work.

7
DAYS LEFT

CHAPTER 4 · SERIK ATA, TOGUZ KUMALAK MASTER

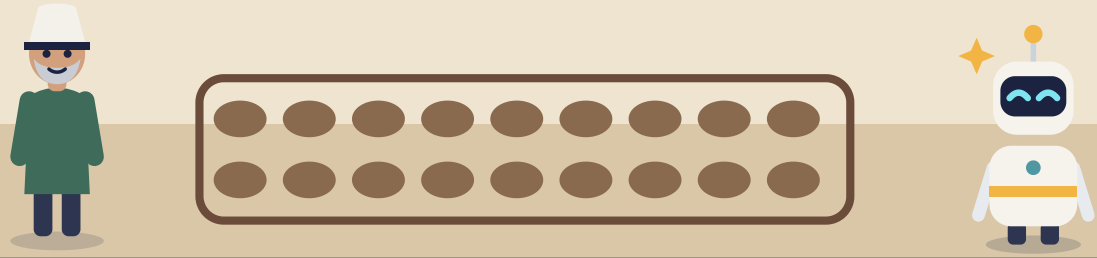
Toguz Kumalak at the Palace

Week 4 · Adversarial search & game trees

Play it
Open this chapter in QAI-QALA

Day 7. Everyone hides from the cold in the Culture Palace.

While it snows, the city plays. Beat my robot at toguz kumalak!



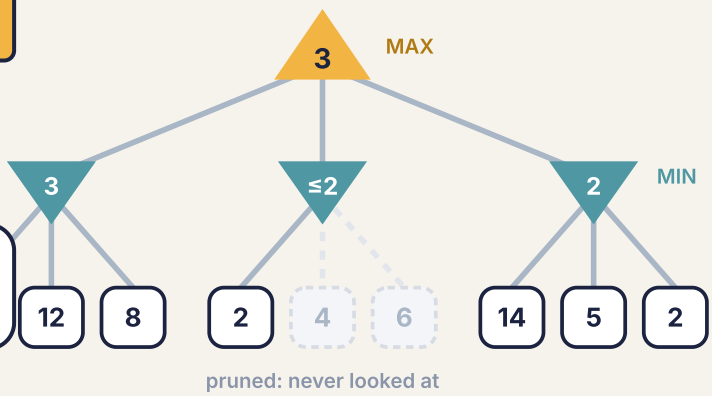
If I move here, the robot answers with the move that hurts me most...



...so for each of my moves I take its MINIMUM. Then I pick the MAXIMUM!



Minimax: MIN values 3, 2, 2 → MAX gets 3.



The middle MIN already shows 2 < 3. Leaves 4 and 6 can't help — skip them!



Alpha–beta! Same answer, less work. You think one move ahead — like a master.



WHAT QADAM LEARNED TODAY

Minimax assumes the opponent plays its best reply. Alpha–beta pruning skips branches that cannot change the decision.



"You think one move ahead — like a true master."

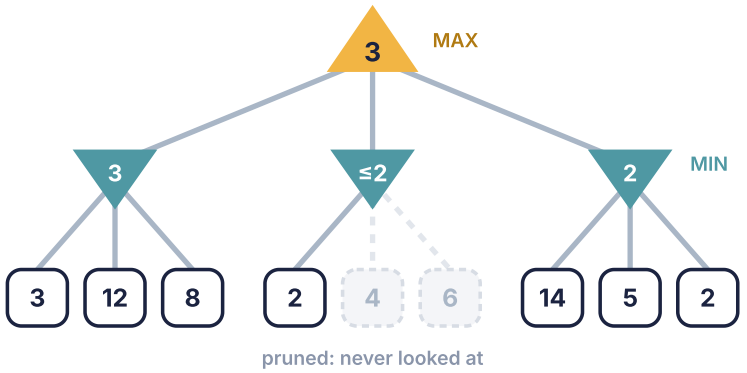
QADAM EXPLAINS · CHAPTER 4

Adversarial search & game trees

Two players, opposite goals, taking turns. MAX (Qadam) maximises the score, MIN (the opponent) minimises it.

KEY IDEAS

Minimax value	MAX node = largest child value, MIN node = smallest. Leaves = game outcomes.
Alpha–beta	α = best already guaranteed for MAX, β = for MIN. Prune when α ≥ β. Same decision, often far fewer nodes.
Expectimax	For dice and luck: chance nodes take the probability-weighted average.



Leaves 3 12 8 | 2 4 6 | 14 5 2. Root = max(3, 2, 2) = 3. Leaves 4 and 6 are pruned.

WORKED EXAMPLE

A gamble: 50% chance of 8 kumalaks, 50% of 2 → expected 0.5·8 + 0.5·2 = **5**. A safe move gives 4. Expectimax picks the gamble (5 > 4).

QADAM'S GLITCH · MYTH VS FACT

"Alpha–beta can change the best move."

Myth. It returns exactly the minimax decision — it only skips work. Good move ordering makes it prune more.

KAZAKH WORD OF THE DAY

Ойын

oiyn — game

DID YOU KNOW?

Toguz kumalak is played on two rows of 9 holes with 162 stones. Deep Blue beat Kasparov at chess in 1997; AlphaGo beat Lee Sedol at Go in 2016.

CHECK YOURSELF

1. What does a MIN node do, and why?
2. Does alpha–beta ever change the minimax decision?
3. 30% chance of 10, otherwise 0. Expected value?

Answers on page 24.

Try it in QAI-QALA. District 4: solve its task with fresh numbers every time — guessing won't work.

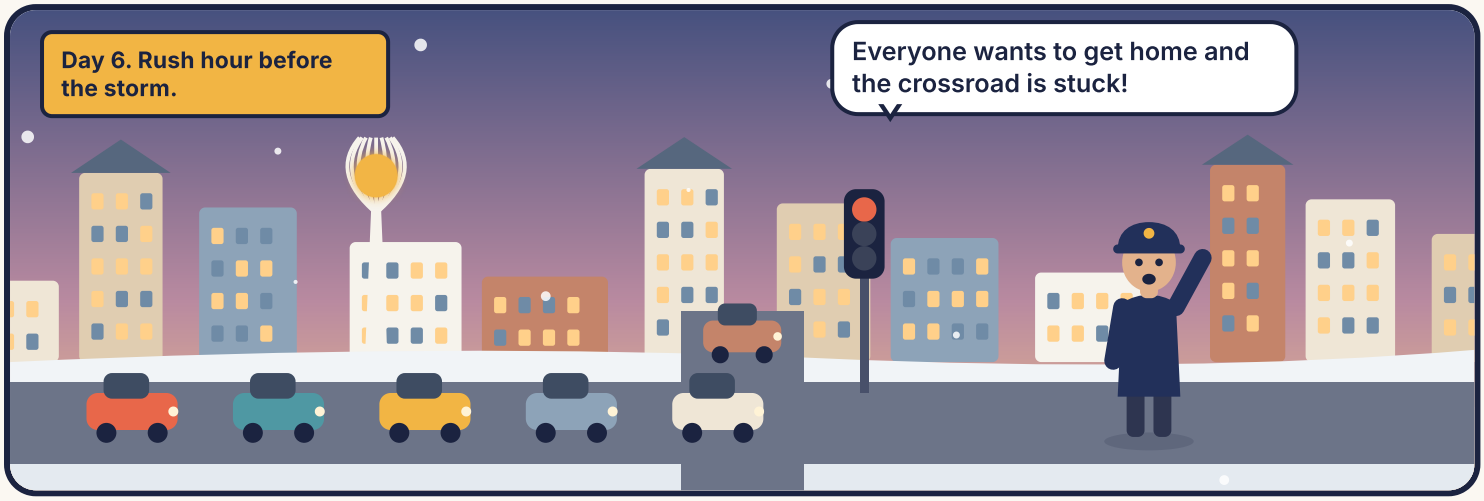
6
DAYS LEFT

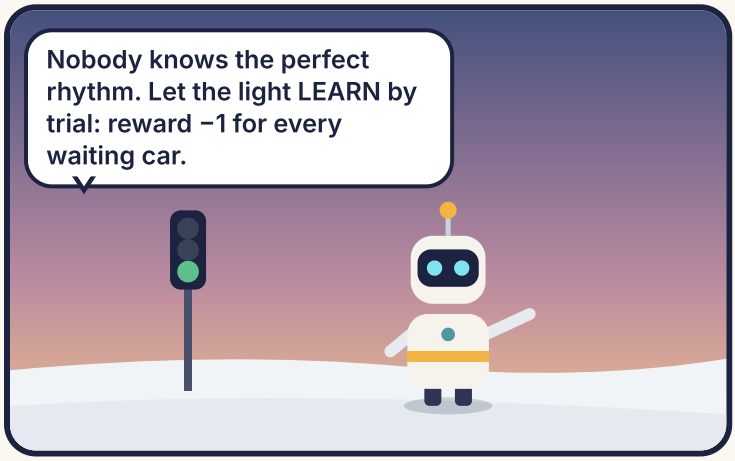
CHAPTER 5 · DANA, TRAFFIC INSPECTOR

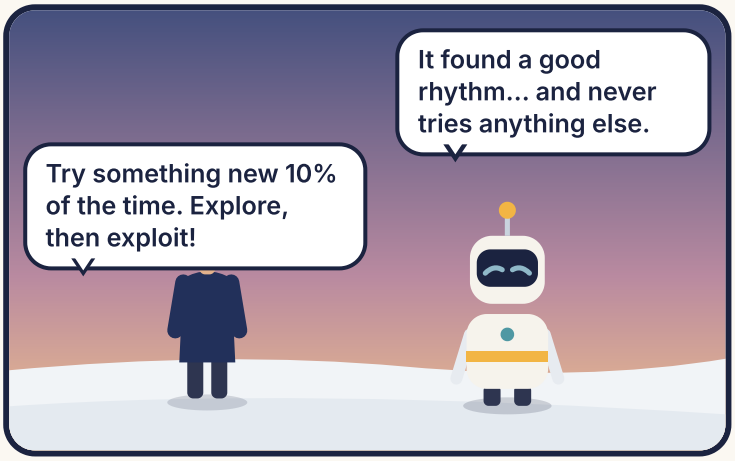
The Jam at the Crossroad

Week 5 · MDPs & reinforcement learning

Play it
Open this chapter in QAI-QALA







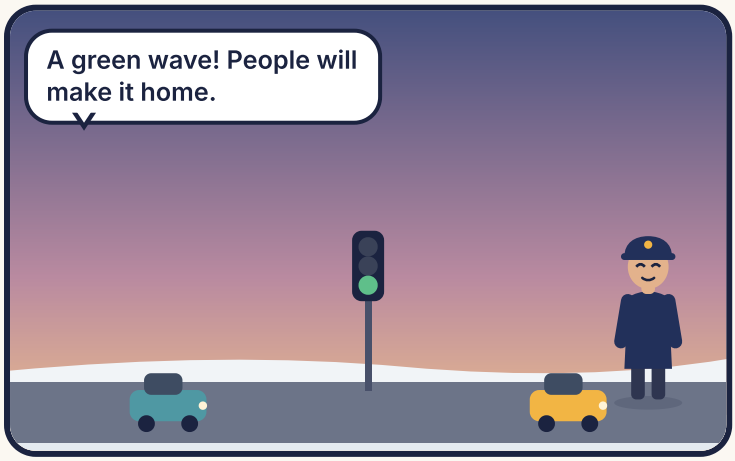
After every switch: update the guess Q.

STATE s
cars waiting N/E

ACTION a
switch / keep

REWARD r
−1 per waiting car

$$Q(s,a) \leftarrow Q(s,a) + \alpha \cdot (r + \gamma \cdot \max Q(s',a') - Q(s,a))$$



WHAT QADAM LEARNED TODAY

An MDP describes acting under uncertainty. Reinforcement learning finds a good policy from rewards alone, balancing exploration and exploitation.

"The jam is gone. People will make it home in time."

QADAM EXPLAINS · CHAPTER 5

MDPs & reinforcement learning

A Markov decision process: states S, actions A, transition model P(s′|s,a), reward R, discount γ. “Markov”: the future depends only on the present state.

KEY IDEAS

Policy π(s) — What to do in every state. The goal: maximise expected discounted reward.

Value iteration — Bellman update: U(s) ← R(s) + γ max_a Σ P(s′|s,a) U(s′), repeated until it settles.

Q-learning — No model needed: learn Q(s,a) from experience. ε-greedy: explore with probability ε.



The learning loop of the traffic light.

WORKED EXAMPLE

Return with γ = 0.5 for rewards 4, 2, 8: G = 4 + 0.5·2 + 0.25·8 = **7**. Q-update, α = γ = 0.5: Q = 2, r = 1, max Q′ = 6 → target 1 + 3 = 4 → Q = 2 + 0.5·(4 − 2) = **3**.

QADAM’S GLITCH · MYTH VS FACT

“RL agents are told the right answer.”

Myth. Nobody gives the correct action — only a reward, often late. That is why exploring matters.

KAZAKH WORD OF THE DAY

Бағдаршам

bağdarşam — traffic light

DID YOU KNOW?

TD-Gammon (Tesauro, 1992) taught itself backgammon to near world-champion level by playing against itself — an early success of reinforcement learning.

CHECK YOURSELF

1. What does “Markov” mean in an MDP?
2. γ = 0.5, rewards 2, 4, 0. What is the return G?
3. Why should a learning agent sometimes explore?

Answers on page 24.

Try it in QAI-QALA. District 5: solve its task with fresh numbers every time — guessing won’t work.

5
DAYS LEFT

CHAPTER 6 · YERLAN, SCHOOL PRINCIPAL

Close the School?

Week 6 · Probability & Bayesian networks

Play it
Open this chapter in QAI-QALA

Day 5. The school.

The pressure sensor beeped. Do I close the school tomorrow?

BEEP!

SCHOOL

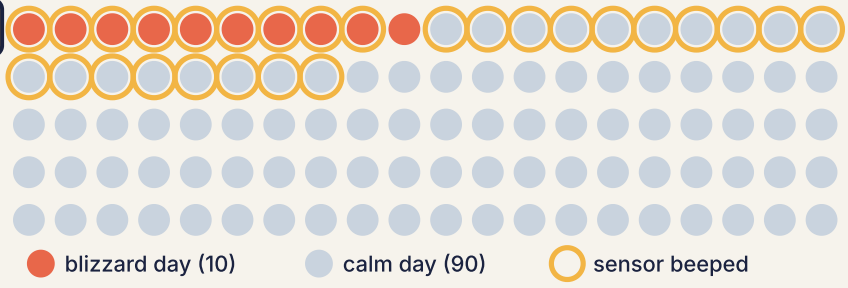
The sensor is right 90% of the time — so it's a 90% blizzard!

Are you sure?

Blizzards come on only 10% of days. And it falsely beeps on 20% of calm days.

Imagine 100 days.

Most beeps are false alarms!



$$P(\text{blizzard} \mid \text{beep}) = 9 / (9 + 18) = 1/3 \approx 33\%$$

Then the wind sensor beeped too: ≈ 69%. Schools close, lessons go online. Numbers, not panic.

WHAT QADAM LEARNED TODAY

Bayes' rule turns "how often does the cause produce the evidence" into "how likely is the cause given the evidence". Never forget the base rate.

"The decision was made by calculation. Parents are calm."

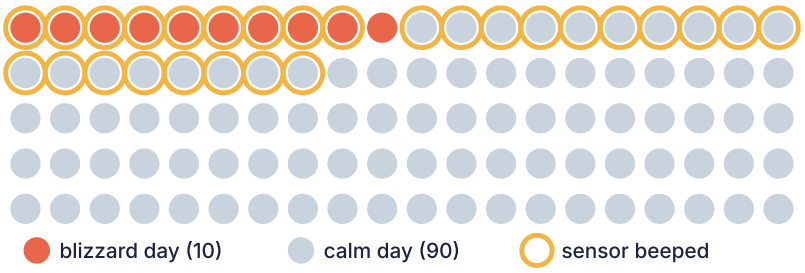
QADAM EXPLAINS · CHAPTER 6

Probability & Bayesian networks

Logic says true or false. Probability says HOW SURE we are — and how to update when evidence arrives.

KEY IDEAS

Conditional probability	P(a | b) = P(a ∧ b) / P(b).
Bayes' rule	P(cause | effect) = P(effect | cause) · P(cause) / P(effect).
Bayesian network	A graph of variables; each node stores P(node | parents). Here: Blizzard → Pressure sensor, Blizzard → Wind sensor.



100 days: 9 real alarms, 18 false alarms.

WORKED EXAMPLE

P(B | beep) = 0.9·0.1 / (0.9·0.1 + 0.2·0.9) = 0.09 / 0.27 = **1/3**.
Each independent beep multiplies the odds by 0.9 / 0.2 = 4.5: odds 1:9 → 1:2 → 9:4, so two beeps give 9/13 ≈ **69%**.

QADAM'S GLITCH · MYTH VS FACT

"A 90%-accurate sensor means a 90% chance."

Myth: base-rate neglect. When the cause is rare, most alarms are false — even from a good sensor.

KAZAKH WORD OF THE DAY

Ықтималдық

yqtimaldyq — probability

DID YOU KNOW?

Thomas Bayes' essay with the rule was published in 1763, two years after his death. Spam filters, medical tests and self-driving cars all rely on it.

CHECK YOURSELF

1. If blizzards came on 50% of days, what would P(blizzard | beep) be?
2. Why do the two sensors become independent once we know whether there is a blizzard?
3. What is a prior?

Answers on page 24.

Try it in QAI-QALA. District 6: solve its task with fresh numbers every time — guessing won't work.

4
DAYS LEFT

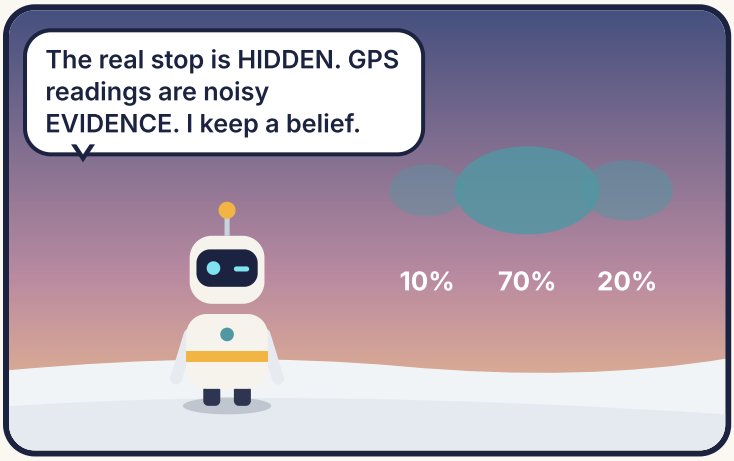
CHAPTER 7 · MARAT, BUS DRIVER

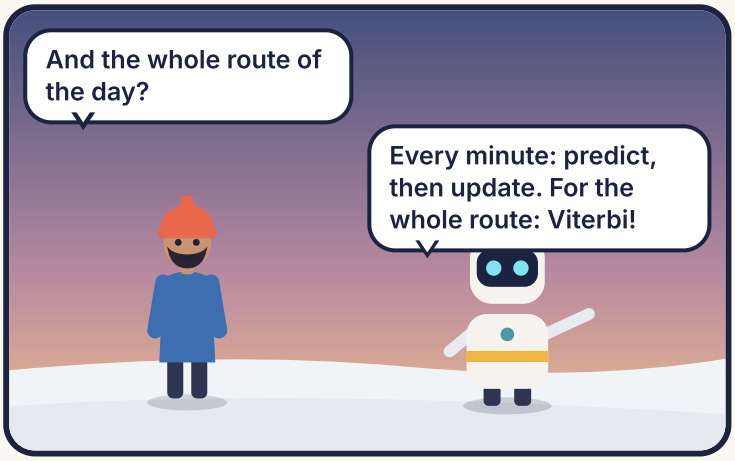
Where Is Bus 12?

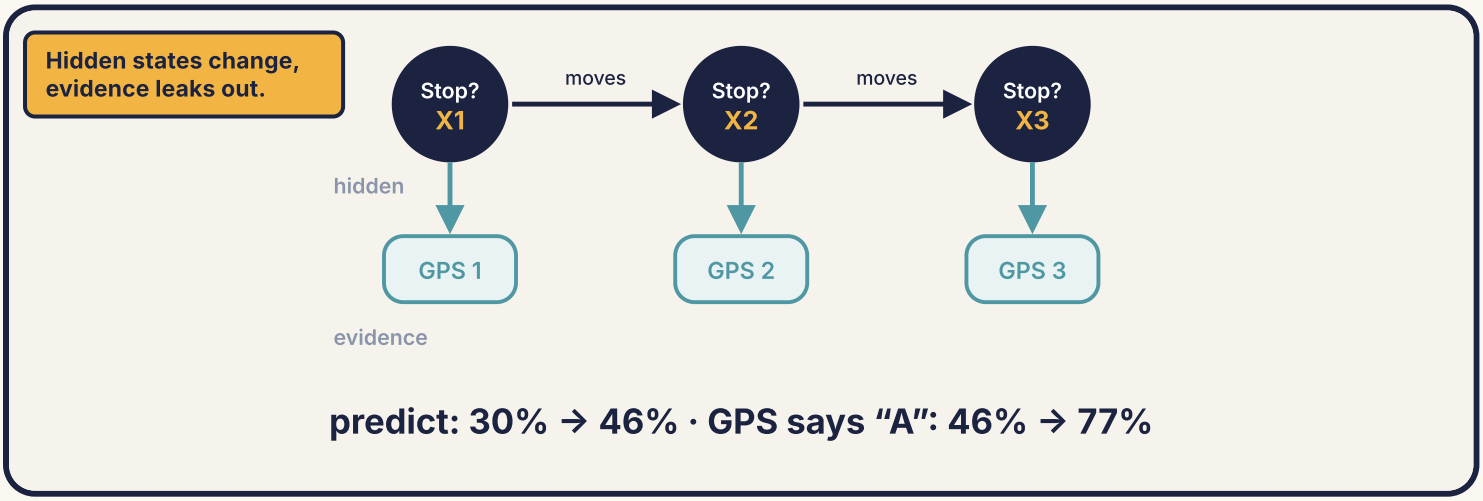
Week 7 · Hidden Markov models & filtering

Play it
Open this chapter in QAI-QALA

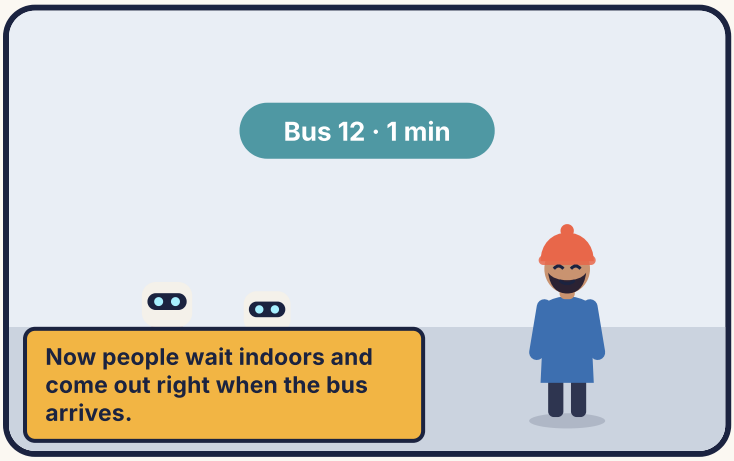








predict: 30% → 46% · GPS says "A": 46% → 77%



WHAT QADAM LEARNED TODAY

An HMM tracks a hidden state over time from noisy evidence: predict with the transition model, update with the sensor model.

"Now people wait indoors and come out right when the bus arrives."

QADAM EXPLAINS · CHAPTER 7

Hidden Markov models & filtering

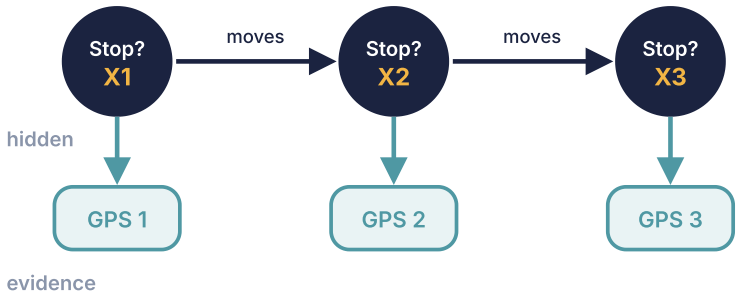
A hidden state changes over time; we only see noisy evidence about it.

KEY IDEAS

Three parts — Prior $P(X_0)$ · transition model $P(X_t \mid X_{t-1})$ · sensor model $P(E_t \mid X_t)$.

Filtering — Belief now = PREDICT (push through transitions) then UPDATE (multiply by evidence, normalise).

Viterbi — The single most likely sequence of hidden states — the bus route of the whole day.



Hidden stops $X_1 \to X_2 \to X_3$, each emitting a GPS reading.

WORKED EXAMPLE

Two stops A, B. P(A) = 30%, the bus stays with 60%.
Predict: 0.3·0.6 + 0.7·0.4 = **46%**.
GPS says A and is right 80% of the time: 0.46·0.8 = 0.368 vs 0.54·0.2 = 0.108 → 0.368 / 0.476 ≈ **77%**.

QADAM'S GLITCH · MYTH VS FACT

"The newest GPS reading is always right."

Myth. One reading is noisy evidence. Filtering combines it with everything we already believed.

KAZAKH WORD OF THE DAY

Аялдама

aialdama — bus stop

DID YOU KNOW?

The Kalman filter (1960) — the continuous cousin of HMM filtering — helped guide the Apollo spacecraft. Phones use the same idea for GPS today.

CHECK YOURSELF

1. Name the three parts of an HMM.
2. Which algorithm gives the most likely whole route?
3. What does the update step do with the evidence?

Answers on page 24.

Try it in QAI-QALA. District 7: solve its task with fresh numbers every time — guessing won't work.

3
DAYS LEFT

CHAPTER 8 · ZHANNA, AKIMAT SECRETARY

Letters and Snowploughs

Week 8 · Rational decisions & classical machine learning

Play it
Open this chapter in QAI-QALA

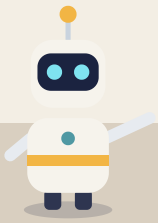
Day 3. The akimat.

Urgent blizzard letters are drowning in spam!



Don't write rules — learn them from labelled examples!

train()
1,000 labelled letters

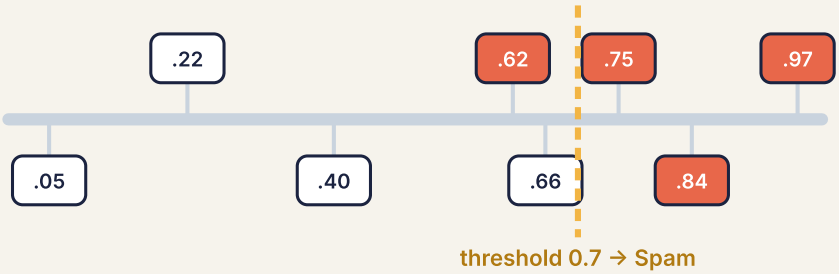


Snow is 40% likely. A plough trip costs 10. Snow damage would cost 50.

0.4 × 50 = 20 > 10. Send it!



A spam score for every letter.



red = real spam · one spam (0.62) slips through, no good letter is lost

Important letters on top. Ploughs out exactly where it pays off.



WHAT QADAM LEARNED TODAY

A rational agent maximises expected utility. Machine learning builds a model from data; a good model generalises to new data, not just the training set.



"Important letters are on top, and the ploughs went out exactly when needed."

QADAM EXPLAINS · CHAPTER 8

Rational decisions & classical machine learning

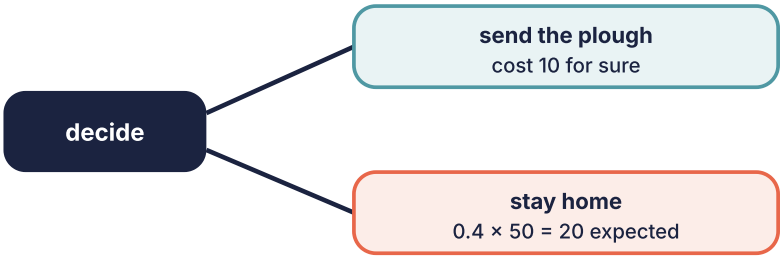
Rational decisions combine beliefs (probabilities) and wants (utilities). Machine learning gets the beliefs from data.

KEY IDEAS

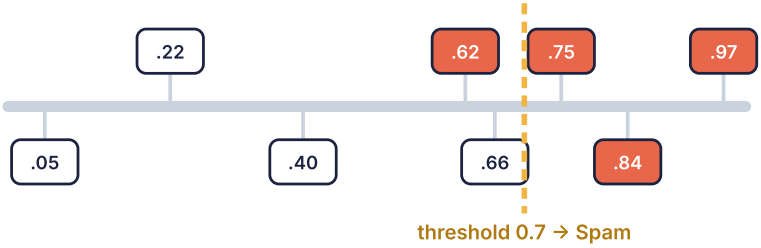
MEU principle — Choose the action with the maximum expected utility: EU(a) = Σ P(outcome | a) · U(outcome).

Types of learning — Supervised (labelled data: spam / not spam) · unsupervised (find groups) · reinforcement (rewards).

Generalisation — Train on one set, test on unseen data. Too complex a model memorises: overfitting.



Expected cost of staying home (20) is higher than the trip (10).



Threshold 0.7: one false negative, zero false positives.

WORKED EXAMPLE

Naive Bayes spam filter: P(spam | words) ∝ P(spam) · Π P(word | spam). "Naive" = words treated as independent given the class — wrong, yet it works well.

QADAM'S GLITCH · MYTH VS FACT

"100% training accuracy means a great model."

Myth. It may just memorise the data (overfitting). Only accuracy on new test data counts.

KAZAKH WORD OF THE DAY

Хат

hat — letter

DID YOU KNOW?

The first mass unsolicited email is usually dated to 1978, sent to ARPANET users. Today most email traffic is filtered by learned classifiers.

CHECK YOURSELF

1. State the MEU principle in one sentence.
2. Raising the spam threshold creates more of which error?
3. Why do we keep a separate test set?

Answers on page 24.

Try it in QAI-QALA. District 8: solve its task with fresh numbers every time — guessing won't work.

2
DAYS LEFT

CHAPTER 9 · TIMUR, POWER-GRID ENGINEER

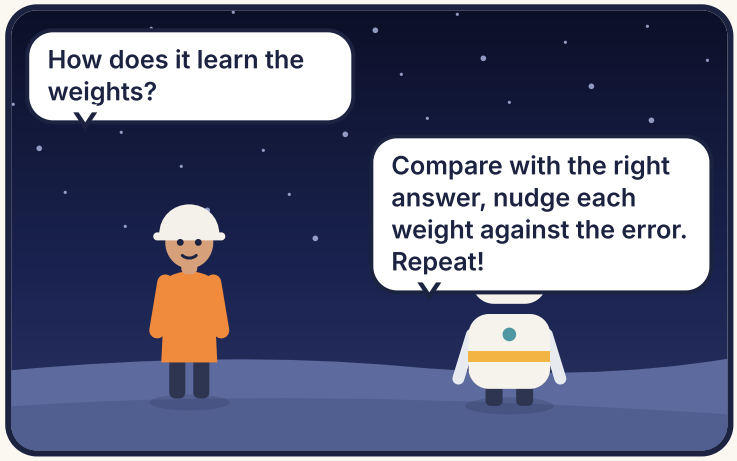
Light Up the City

Week 9 · Perceptrons & neural networks

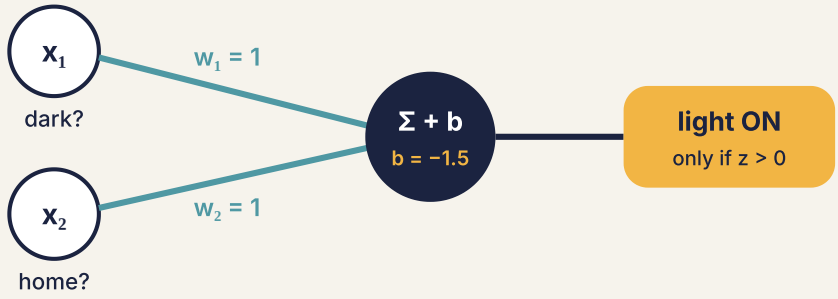
Play it
Open this chapter in QAI-QALA







A perceptron.



$$z = 1 \cdot x_1 + 1 \cdot x_2 - 1.5 > 0 \text{ only when BOTH are 1} \rightarrow \text{logical AND}$$



WHAT QADAM LEARNED TODAY

A neuron computes a weighted sum plus a bias and fires through an activation. Networks learn by gradient descent; backpropagation finds the gradients.

“Windows glow where people are, and power is saved where nobody is home.”

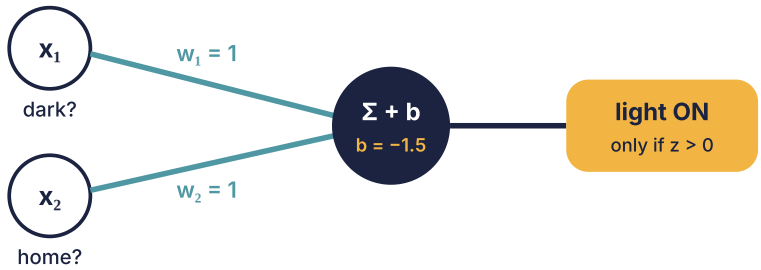
QADAM EXPLAINS · CHAPTER 9

Perceptrons & neural networks

From one artificial neuron to deep learning.

KEY IDEAS

Neuron	$z = w_1x_1 + w_2x_2 + \ldots + b$, output = activation(z): step, sigmoid or ReLU.
Perceptron rule	After a mistake: $w \leftarrow w + \alpha \cdot (y - \hat{y}) \cdot x$. Learns AND and OR, but never XOR (not linearly separable).
Backpropagation	Hidden layers solve XOR. The chain rule gives every weight's gradient; gradient descent steps against it.



The window neuron: weights 1 and 1, bias −1.5 → AND.

WORKED EXAMPLE

Gradient descent, learning rate $\eta = 0.1$: $w = 0.5$, gradient 2 → $w = 0.5 - 0.1 \cdot 2 =$ **0.3**. A negative gradient (−3) would raise it: 0.5 + 0.3 = **0.8**.

QADAM'S GLITCH · MYTH VS FACT

"Neural networks work just like the human brain."

Myth. They are loosely inspired by neurons, but learn with maths (gradients) the brain does not use in this way.

KAZAKH WORD OF THE DAY

Жарық

jaryq — light

DID YOU KNOW?

Rosenblatt built the perceptron in 1958. Backpropagation became famous with Rumelhart, Hinton and Williams in 1986 — the base of today's deep learning.

CHECK YOURSELF

1. Inputs x = (1, 0), weights (1, 1), bias −1.5. Does the window light?
2. Why can one perceptron not learn XOR?
3. What does backpropagation compute?

Answers on page 24.

Try it in QAI-QALA. District 9: solve its task with fresh numbers every time — guessing won't work.

1
DAY LEFT

CHAPTER 10 · ALIYA, SCHOOLGIRL

An Honest Assistant

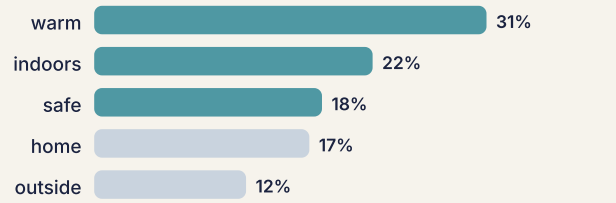
Week 10 · LLMs, generative AI & AI safety

Play it
Open this chapter in QAI-QALA



An LLM predicts the NEXT TOKEN, again and again.

"During the blizzard, please stay ..."



top-p 60% keeps the teal words



Ground answers in sources. Protect privacy.

Where can I warm up?

Culture Palace, 5 minutes on foot.

When was Bayterek built?

It opened in 2002. I checked the city source.

Give me my neighbour's address.

Sorry, I don't share personal data.



WHAT QADAM LEARNED TODAY

LLMs generate text by predicting the next token with a transformer. They can hallucinate, so facts must be checked, privacy protected and AI use disclosed.

"The assistant answers precisely, admits what it doesn't know and keeps secrets safe."

QADAM EXPLAINS · CHAPTER 10

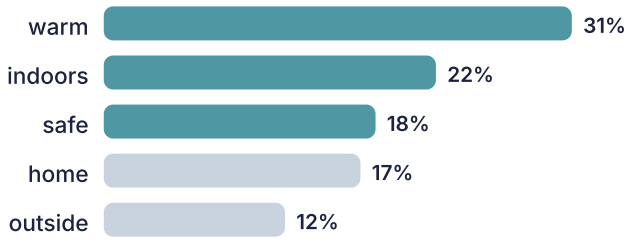
LLMs, generative AI & AI safety

How chat assistants work — and how to use them responsibly.

KEY IDEAS

Tokens & embeddings	Text is split into tokens; each becomes a vector of numbers.
Transformer	Self-attention lets every token weigh every other: query · key decides how much of each value to take.
Training	Pre-train on next-token prediction, then fine-tune (e.g. with human feedback) to be helpful and safe.

"During the blizzard, please stay ..."



Next-token probabilities; top-p 60% keeps 31 + 22 + 18 = 71%.

SAFETY CHECKLIST

- **Hallucination**: confident but false — verify with sources (retrieval helps).
- **Privacy**: never reveal personal data.
- **Bias**: models learn the biases of their data.
- **Honesty**: disclose when AI helped you.

WORKED EXAMPLE

Temperature 0 always takes the most likely token ("warm"); higher temperature spreads the choice. Top-k = 2 keeps 31 + 22 = **53%** of the probability.

QADAM'S GLITCH · MYTH VS FACT

"AI understands and feels like a human."

Myth. An LLM is trained to predict text. Fluent language is not understanding or feelings.

KAZAKH WORD OF THE DAY

Көмекші

kömekşi — assistant, helper

DID YOU KNOW?

The transformer was introduced in "Attention Is All You Need" (Vaswani et al., 2017). It powers almost every modern chat assistant.

CHECK YOURSELF

1. What does an LLM predict at every step?
2. What is a hallucination?
3. Top-p 60% for 40%, 25%, 20%, 15%: how many words stay?

Answers on page 24.

Try it in QAI-QALA. District 10: solve its task with fresh numbers every time — guessing won't work.

0
DAYS LEFT

FINALE

The Night of the Blizzard

Wind 25 m/s, visibility 50 m. Ten systems, ten lessons — working together.



1. The weather station warned the whole city two hours early.

2. Ambulances took the cheapest routes through the snow.

3. Buses kept running on lines that never collide.

4. The Culture Palace stayed warm and full all night.

5. Traffic lights cleared the roads in minutes.

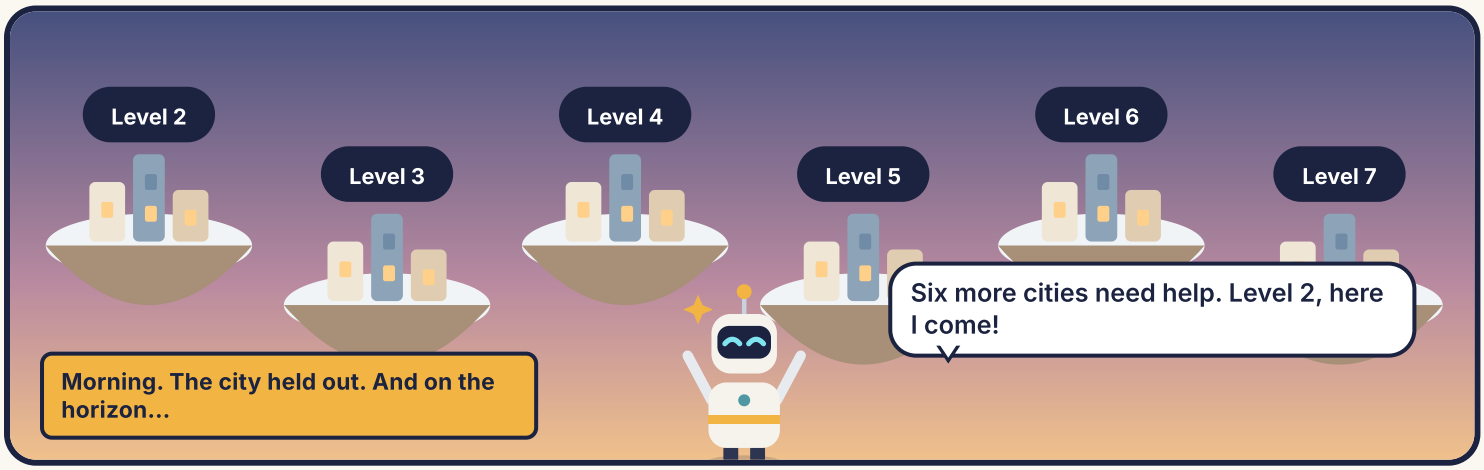
6. Schools closed in time, with no false alarms all week.

7. Every passenger knew where their bus was.

8. Snowploughs went out where the expected damage was highest.

9. The grid held: lights on in every home with people in it.

10. The Bayterek assistant answered thousands of questions honestly.



To be continued in QAI-QALA. Scan to play: build all ten districts, face the blizzard night yourself and see what waits on the six islands.

ANSWERS

Check yourself — solutions

CHAPTER 1 · THE ROBOT WHO COULDN'T SEE

1. Which PEAS letter is the snow gauge?
   S — a sensor.
2. Is "IF snow ≥ 6 cm THEN plough" a reflex or a goal-based agent?
   Simple reflex: it reacts to the current percept only.
3. Qadam cleared 6 paths in 25 steps. What is his score?
   10 × 6 − 25 = 35.

CHAPTER 2 · MEDICINE BEFORE THE STORM

1. Which data structure holds the BFS frontier?
   A FIFO queue.
2. Which algorithm ignores road costs when it chooses?
   Greedy best-first search (it uses only h).
3. True cost from E to the hospital is 5. Is h(E) = 4 admissible?
   Yes: 4 ≤ 5, it never overestimates.

CHAPTER 3 · HOMES AND BUS LINES

1. Values 2, 6, 4, 9. Hill climbing starts at 2. Where does it stop?
   At 6 (its neighbours 2 and 4 are lower).
2. Name the three parts of a CSP.
   Variables, domains, constraints.
3. What does the MRV heuristic choose first?
   The variable with the fewest legal values left.

CHAPTER 4 · TOGUZ KUMALAK AT THE PALACE

1. What does a MIN node do, and why?
   Takes the smallest child value: we assume the opponent plays its best reply.
2. Does alpha–beta ever change the minimax decision?
   No — it only prunes branches that cannot matter.
3. 30% chance of 10, otherwise 0. Expected value?
   0.3 × 10 = 3.

CHAPTER 5 · THE JAM AT THE CROSSROAD

1. What does "Markov" mean in an MDP?
   The next state depends only on the current state and action.
2. γ = 0.5, rewards 2, 4, 0. What is the return G?
   2 + 0.5·4 + 0.25·0 = 4.
3. Why should a learning agent sometimes explore?
   An even better action may exist that it has never tried.

CHAPTER 6 · CLOSE THE SCHOOL?

1. If blizzards came on 50% of days, what would P(blizzard | beep) be?
   0.45 / (0.45 + 0.10) ≈ 82%.
2. Why do the two sensors become independent once we know whether there is a blizzard?
   Both are caused only by the blizzard: conditional independence given the parent.
3. What is a prior?
   The belief before seeing the evidence, e.g. P(blizzard) = 10%.

CHAPTER 7 · WHERE IS BUS 12?

1. Name the three parts of an HMM.
   Prior, transition model, sensor model.
2. Which algorithm gives the most likely whole route?
   The Viterbi algorithm.
3. What does the update step do with the evidence?
   Multiplies each state by how well it explains the evidence, then normalises.

CHAPTER 8 · LETTERS AND SNOWPLOUGHS

1. State the MEU principle in one sentence.
   Pick the action whose probability-weighted utility is highest.
2. Raising the spam threshold creates more of which error?
   More false negatives (spam slips into the inbox).
3. Why do we keep a separate test set?
   To measure how well the model generalises to unseen data.

CHAPTER 9 · LIGHT UP THE CITY

1. Inputs x = (1, 0), weights (1, 1), bias −1.5. Does the window light?
   z = 1 − 1.5 = −0.5 ≤ 0 → no.
2. Why can one perceptron not learn XOR?
   XOR is not linearly separable — no single straight line splits it.
3. What does backpropagation compute?
   The gradient of the loss with respect to every weight (chain rule).

CHAPTER 10 · AN HONEST ASSISTANT

1. What does an LLM predict at every step?
   The next token.
2. What is a hallucination?
   A fluent, confident answer that is false.
3. Top-p 60% for 40%, 25%, 20%, 15%: how many words stay?
   2 (40 + 25 = 65% ≥ 60%).

BACK MATTER

Glossary · References · AI use · Team

GLOSSARY

Agent — perceives through sensors, acts through actuators
Rational agent — maximises expected performance
PEAS — performance, environment, actuators, sensors
BFS / DFS — FIFO queue / LIFO stack search
UCS — expands the cheapest path g
A* — expands smallest f = g + h
Admissible h — never overestimates
Hill climbing — move to the best neighbour
Annealing — accept worse moves with e^(ΔE/T)
CSP — variables, domains, constraints
Minimax — MAX of MINs
Alpha–beta — prunes branches that cannot matter
Expectimax — average over chance nodes
MDP — states, actions, P(s'|s,a), rewards, γ
Q-learning — model-free learning of Q(s,a)
Bayes' rule — P(A|B) = P(B|A)P(A)/P(B)
Bayesian network — graph of conditional probabilities
HMM — hidden states + noisy evidence
Filtering — predict, then update
Viterbi — most likely hidden sequence
MEU — maximise expected utility
Overfitting — memorises training data
Perceptron — weighted sum + step
Backpropagation — chain-rule gradients
Token — piece of text an LLM predicts
Attention — weighs every token against every other
Hallucination — fluent but false output

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AI USE DECLARATION

We used **Claude (Anthropic)** to help draft text, generate the code that draws the illustrations and diagrams, and build the QAI-QALA game. We designed the story, characters and structure, checked every explanation against the textbook, and recomputed every number (search results, minimax, Bayes, HMM, Q-updates) with our own code. Mistakes that remain are ours.

TEAM 11

Yuldashov Bilal — game design, 3D city, chapters 1–5, testing all algorithms.
Tashmetov Shakhrukh — chapters 6–10, quizzes and myths, presentation card.
Both — the comic story, checking every example, presentation and defense.

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AI HISTORY IN TEN STEPS

1950 — Turing asks "Can machines think?" and proposes the imitation game

1956 — Dartmouth workshop: the field gets its name

1958 — Rosenblatt's perceptron

1968 — A* search, built for the robot Shakey

1986 — Backpropagation trains multi-layer networks

1997 — Deep Blue beats Kasparov at chess

2012 — AlexNet starts the deep-learning boom

2016 — AlphaGo beats Lee Sedol at Go

2017 — The transformer: "Attention Is All You Need"

2022 — ChatGPT brings LLMs to everyone



"I learned to sense, search, plan, play, learn, doubt, track, decide, think and talk — one step a day. **Qadam** means step. Now it's your turn: open QAI-QALA and build the city yourself."